1.1.9. Breadth Fir… 08:13

Write a program to implement Breadth First Search (BFS) graph traversal methods.

Input Format:

- The first line contains an integer N, the number of vertices in the graph.
- The second line contains an integer E, the number of edges in the graph.
- The next E pairs of lines contain two integers each:
  - The first line of each pair contains an integer s representing the source vertex.
  - The second line of each pair contains an integer d representing the destination vertex.
- The last line contains an integer v, which is the starting vertex for the BFS traversal.

Output Format:

- The program will output the vertices of the graph, each on a new line, in the order they are visited by the BFS traversal, starting from the vertex v.

Note:

- Driver code is already provided to you in the editor, your task is to fill in the required code.
- Refer to visible test cases for better understanding.

Sample Test Cases +

Explorer

C GraphsB… Submit

```c
#include <stdio.h>
#include <stdlib.h>
#define MAX 99

struct node {
    int vertex;
    struct node* next;
};
typedef struct node* GNODE;

GNODE graph[20];
int visited[20];
int queue[MAX], front = -1, rear = -1;
int n;

void insertQueue(int vertex) {
    if (rear == MAX - 1)
        printf("Queue Overflow.\n");
    else {
        if (front == -1)
            front = 0;
        rear++;
        queue[rear] = vertex;
    }
}

int isEmptyQueue() {
    return (front == -1 || front > rear);
}

int deleteQueue() {
    if (isEmptyQueue()) {
```

Debugger

‹ Prev Reset Submit Next ›

```
        printf("Queue Underflow\n");
        exit(1);
    }
    return queue[front++];
}

void BFS(int v) {
    insertQueue(v);
    visited[v] = 1;

    while (!isEmptyQueue()) {
        int curr = deleteQueue();
        printf("\n%d", curr);

        GNODE p = graph[curr];
        while (p != NULL) {
            int neighbor = p->vertex;
            if (!visited[neighbor]) {

insertQueue(neighbor);

visited[neighbor] = 1;
            }
            p = p->next;
        }
    }
```

```
66
67  void main() {
        int N, E, s, d, i,
68  v;
69      GNODE p, q;
70
        printf("Enter the
71  number of vertices: ");
72      scanf("%d", &N);
73      n = N;
74
        for (i = 0; i < N;
75  i++) {
            graph[i] =
76  NULL;
77          visited[i] = 0;
78      }
79
        printf("Enter the
80  number of edges: ");
81      scanf("%d", &E);
        for (i = 1; i <=
82  E; i++) {
            printf("Enter
83  source: ");
            scanf("%d",
84  &s);
            printf("Enter
85  destination: ");
            scanf("%d",
86  &d);
87
            q =
        (GNODE)malloc(sizeof(st
88  ruct node));
89          q->vertex = d;
90          q->next = NULL;
91
            if (graph[s]
92  == NULL) {
                graph[s] =
93  q;
94          } else {
```